

Meaning of Research

Research is a systematic and scientific process of searching for new knowledge or finding solutions to problems through data collection, analysis, and interpretation.

Objectives of Research

- To discover new facts
- To gain knowledge and understanding
- To describe situations accurately
- To test relationships between variables

Motivation in Research

People conduct research due to curiosity, desire to solve problems, academic requirements, and the need to contribute to society.

Types of Research

- Applied Research:
Research that focuses on solving a real-life or practical problem.
- Fundamental (Basic) Research:
Research that aims to increase knowledge or develop theories, without immediate practical use.
- Quantitative Research:
Research that uses numerical data and statistics to analyze results.
- Qualitative Research:
Research that focuses on opinions, behaviors, and experiences, not numbers.
- Conceptual Research:
Research based on ideas, concepts, or theories, without direct observation.
- Empirical Research:
Research based on observation or experiment, using real-world

Research Approaches

- Quantitative: Uses numerical data and statistics
- Qualitative: Focuses on opinions, behaviors, and experiences

Significance of Research

- Research helps in solving problems
- improving decision-making,
- supporting development in society and technology.

Research Methods vs Methodology

Research methods are tools used to collect data, **while** research methodology is the overall strategy and reasoning behind choosing those methods.

Research Process

- *Identify research problem
- *Review literature
- * Form hypothesis
- * Collect data
- * Analyze data
- * Present results

Criteria of Good Research

- Good research should be systematic
- Good research should be logical,
- Good research should be Objective
- Good research should be accurate and reliable.

REFLECTION QUESTIONS

1.How would you explain concept of research in your own word

Research is a structured way of investigating a problem in order to find accurate answers using scientific methods.

2. Why is it important for university student to understand research method

It helps students develop critical thinking, problem-solving skills, and the ability to evaluate information effectively.

3. Which type of research do you find most relevant to your field of study and why

Analytical research is most relevant in my field (IT/Programming) because it helps analyze systems, detect problems, and improve solutions.

4. What challenge might beginner face when conducting research

Beginners may face challenges such as lack of experience, difficulty in choosing a topic, data collection problems, and understanding research methods.

5. how can research contributing to solving real life problem in your community

Research helps solve real-life problems by providing practical solutions, such as improving local businesses, developing technology systems, and addressing community challenges.